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| **Fiscal Year:** | FY 2016 | **Task Last Updated:** | FY 09/21/2015 |
| **PI Name:** | Karma, Alain Ph.D. | | |
| **Project Title:** | Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Materials science | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 02115-5005 | **Congressional District:** | 8 |
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| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA MSFC |
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| **Flight Program:** | | | |
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| **Key Personnel Changes/Previous PI:** | | | |
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| **Performance Goal Text:** | | | |
| | Cellular and dendritic microstructures formed during industrial solidification processes (such as casting, welding, and brazing) determine the properties of commercial metallic alloys produced in numerous industries, with applications ranging from automobile engine parts to aircraft turbine blades. The production of innovative materials and the optimization of both properties and processing routes go through an accurate prediction and modeling of the dynamics of microstructures formation. From a fundamental standpoint, the formation of these structures is controlled by a subtle interplay between microscopic interfacial phenomena and macroscopic transport of heat and solute, which is strongly influenced by gravity-induced melt convection under terrestrial conditions. Therefore, microgravity experiments provide a unique means to investigate microstructure development with negligible convection and to test the predictions of theoretical models in a purely diffusive growth regime. | | |

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| **Task Description:** | While most of the theoretical research to date has focused on the growth of single crystals, most commercially relevant metallic alloys are polycrystalline. The main aim of this research is to bring a quantum leap in our ability to model realistically dendritic alloy structures in polycrystalline materials where interactions between well-developed dendrites from the same grain and from grains of different crystal orientations both need to be considered. Interactions at grain boundaries are especially important because of their influence on dendrite fragmentation and the formation of solidification defects.<br><br>The main goal of this project is to develop state-of-the-art computational tools to model dendritic microstructures formation under diffusive growth conditions and to establish benchmark comparisons with experimental observations under microgravity conditions. Those comparisons have the dual purpose to obtain new fundamental insights into microstructure formation and to validate computational models that can ultimately be used in an industrial setting to guide the design of structural alloys and optimize solidification processing conditions. Computational tools include the phase-field methods and a new dendritic needle network (DNN) model. The phase-field method resolves the details of the dendrite tip scale and is ideally suited to model cellular and dendritic structures in dilute alloys on a scale comparable to the primary spacing. The DNN model is designed to model dendritic microstructure formation on a grain scale that is orders of magnitude larger than the dendrite tip scale. Unlike previous models of grain structure evolution, the DNN approach models explicitly the growth kinetics of all branches of the hierarchical dendritic network (including primary, secondary, and tertiary branches) that interact diffusively via the solutal field. Hence, this method is ideally suited to investigate complex mechanisms of dendritic microstructure formation including the competitive growth of misoriented columnar grains and the columnar-to-equiaxed transition. This transition is of practical relevance for the casting industry and is being investigated experimentally aboard the International Space Station (ISS) in collaboration with our European partners in the framework of the CETSOL project. Simulation results are critically important to interpret the results of those experiments in both metallic and transparent organic alloys and to guide future experiments.<br><br>ED. NOTE (12/14/2015): Project continues as "Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys--NNX16AB54G." See that project for subsequent reporting. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Many commercial structural alloys, from nickel based superalloys used in turbine blades to light weight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. This research is aimed at developing computational tools to predict reliably the formation of these dendritic microstructures on experimentally relevant length and time scales so as to accelerate and optimize the design of new technological alloys that can directly benefit US industries. |
| **Task Progress:** | During the current funding period, we completed the implementation of the DNN model to describe equiaxed grains of different orientations in two dimensions. This allowed us to use this model to simulate for the first time the columnar-to-equiaxed transition (CET) in a detailed self-consistent computational model that takes into account the transient growth dynamics of randomly oriented equiaxed grains and their solutal interactions with pre-existent columnar grains. The results were compared to observations of the CET in Al-7%wtSi samples solidified aboard the ISS, and used to critically test the predictions of approximate analytical theories of the CET. This comparison revealed that analytical models overestimate the growth rate of equiaxed grains. They also showed that those models provide a reasonable estimate of the CET as a function of temperature gradient and pulling velocity, but do not predict well the occurrence of the CET as a function of nucleant concentration. DNN simulations predict that the CET persists for much smaller nucleant concentration than predicted by simplified models due to the slow overgrowth of the misoriented grains by the columnar front. This slow overgrowth gives rise to a progressive CET that is experimentally observed in Al-7%wtSi alloys. This transition originates from the competitive growth of pre-existing columnar grains and elongated equiaxed grains.<br>We finalized the development of the DNN model for columnar grains in 3D and carried out 3D simulations of directional solidification of Al-7%wtSi samples. The primary spacing was compared quantitatively to measured spacings in Al-7%wtSi samples directionally solidified aboard the ISS. The remarkable quantitative agreement between theory and experiment for the primary dendrite spacing of columnar microstructures in Al-7%wtSi alloys provides an important validation of the DNN model in 3D, showing that this approach is capable to predict key microstructural features on experimentally relevant length and time scales in 3D. The next major step in this project is to complete the implementation of the DNN model in 3D for randomly oriented grains so as to model quantitatively the CET in both metallic and transparent alloy systems investigated in the CETSOL project.<br><br>ED. NOTE (12/14/2015): Project continues as "Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys--NNX16AB54G." See that project for subsequent reporting. |
| **Bibliography Type:** | Description: (Last Updated: 05/10/2026) |
| **Articles in Peer-reviewed Journals** | Tourret D, Karma A. "Growth competition of columnar dendritic grains: A phase-field study." Acta Materialia. 2015 Jan;82:64-83. http://dx.doi.org/10.1016/j.actamat.2014.08.049 , Jan-2015 |
| **Articles in Peer-reviewed Journals** | Tourret D, Clarke AJ, Imhoff SD, Gibbs PJ, Gibbs JW, Karma A. "Three-dimensional multiscale modeling of dendritic spacing selection during Al-Si directional solidification." JOM (Journal of The Minerals, Metals & Materials Society (TMS). 2015 Aug;67(8):1776-85. http://dx.doi.org/10.1007/s11837-015-1444-2 , Aug-2015 |
| **Articles in Peer-reviewed Journals** | Tourret D, Karma A, Clarke AJ, Gibbs PJ, Imhoff SD. "Three-dimensional Dendritic Needle Network model with application to Al-Cu directional solidification experiments." IOP Conference Series: Materials Science and Engineering. 2015 Jun;84:012082. (MCWASP XIV: International Conference on Modelling of Casting, Welding and Advanced Solidification Processes, Awaji island, Hyogo, Japan, June 21–26, 2015.) http://dx.doi.org/10.1088/1757-899X/84/1/012082 , Jun-2015 |